

Skills

- Python
- SQL
- HTML
- CSS
- Data Science
- Machine Learning
- Generative AI
- Microsoft Excel
- Flask
- Docker
- AWS EC2
- Zapier

Projects

CRIME PATTERN ANALYSIS March 2025

- Analyzed Mumbai’s crime trends (2020–2024) across IPC, NDPS, and women-related offenses to uncover actionable patterns.
- Cleaned and structured multi-source crime data using Microsoft Excel and performed EDA using Python (Pandas, Numpy).
- Visualized seasonal crime surges, detection efficiency, and category-wise trends using Matplotlib and Seaborn.

PRODUCT MANAGEMENT CHATBOT January 2024

- Designed to help product managers retrieve product sales insights by converting natural language queries into SQL.
- Used LLaMA 2 to translate user questions into executable SQL queries on the company’s database.
- Applied few-shot learning for complex queries using a custom dataset of 100 Q&A SQL pairs stored in ChromaDB.
- Integrated vector-based retrieval with LLaMA 2 to improve accuracy and handle complex SQL generation.

AI-DRIVEN EDUCATIONAL CHATBOT November 2023

- Developed to help users get accurate, conversational answers to course-related queries using AI.
- Collected a Q&A dataset and created embeddings using Hugging Face models.
- Stored embeddings in a FAISS vector database and retrieved matches using cosine similarity with RetrievalQA.
- Provided human-like responses by passing query context to Google PaLM 2 for final answer generation.

Publications

LFGFORMER: A THEORETICAL ARCHITECTURE FOR LIGHTWEIGHT FUSION OF GRAPH AND TRANSFORMER REPRESENTATIONS

- Proposed LFGFormer, a hybrid model combining transformers and shallow graph networks for efficient multi-modal learning.
- Designed for CPU-friendly environments, enabling fusion of text and structured/tabular data using a lightweight attention-based architecture.
- Applicable in real-world tasks like research paper classification, citation prediction, and financial fraud detection.

ENHANCED STRUCTURAL HEALTH MONITORING THROUGH LSTM-ENHANCED GRADIENT BOOSTING REGRESSOR

- Proposed a novel hybrid model (LE-GBR) combining Gradient Boosting and LSTM to predict infrastructure failure.
- Outperformed traditional models in forecasting structural health using real-world sensor data from bridges.
- Achieved near-perfect prediction accuracy ($R^2 = 0.999999$) with minimal error metrics.

Work Experience

AI AUTOMATION INTERN – HOLISTIQUE BEAUTY PRODUCTS PVT LTD

February 2025 - March 2025

- Built a price scraping tool using Flask, HTML/CSS, and Docker to fetch real-time product prices from platforms like Amazon, Flipkart, and Nykaa; deployed on Amazon EC2.
- Solved scraping failures post-deployment by containerizing the app with Docker, improving stability and response time.
- Designed and implemented a duplicate email detection system using Zapier to prevent repeated customer queries from being reprocessed in Salesforce.
- Automated email replies to only “new” queries while ignoring duplicates, reducing customer support load and improving response efficiency.

DATA SCIENTIST INTERN – EVOASTRA VENTURES

June 2024 - September 2024

- Built a web scraping tool using BeautifulSoup to extract detailed car information from a car-selling platform, including names, fuel types, transmissions, colors, prices, and location-specific on-road prices.
- Cleaned and structured the data into CSV files to enhance accessibility and enable smooth analysis.
- Analyzed company sales data to identify patterns and deliver insights for refining business strategies.
- Preprocessed and explored HR data to support data-driven decisions in workforce planning and management.

MACHINE LEARNING INTERN – HINDUSTAN PETROLEUM CORPORATION LIMITED

July 2023 - August 2023

- Developed predictive models to estimate the quality values of petroleum oil byproducts including Naphtha, 1SS, and ATF oil using real-time refinery process data.
- Achieved 96% accuracy for Naphtha using Support Vector Regressor, 98% for 1SS with Gradient Boosting, and 98% for ATF with Random Forest.
- Applied feature engineering, hyperparameter tuning, and cross-validation to improve model performance and reliability.

Education

BTECH (COMPUTER ENGINEERING) – DJ Sanghvi College Of Engineering – Mumbai, India

2021 - 2025